

E-Commerce Sales & Data Quality Dashboard

Project Domain: E-Commerce Analytics

Project Type: End-to-End Data Analytics Dashboard

Technologies Used: MySQL, Power BI

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Technical Stack:

- **MySQL** – Data validation, cleaning, and KPI computation using 20+ analytical queries
- **Power BI** – DAX-based KPI dashboards, time-series analysis, and interactive slicers

Project Overview:

Developed an end-to-end e-commerce sales analytics solution by cleaning and validating 100K+ records using MySQL and building an interactive Power BI dashboard. The project analyzes revenue trends, product performance, payment behavior, country-wise sales, and order lifecycle impact to generate actionable business insights.

Business Problem Statement:

The dataset contained inconsistencies in city–country and product–category mappings, resulting in inaccurate reporting. The objective was to clean, validate, and standardize the data to enable reliable business insights.

Project Objectives:

- Perform data validation and correction in MySQL to ensure accurate location and product classification

- Compute key KPIs including revenue, orders, average order value (AOV), customers, and valid order percentage
- Analyze revenue impact across product categories, payment methods, countries, and order lifecycle
- Develop an interactive Power BI dashboard using DAX and slicers for business reporting

Project Outcomes:

Achieved 70.1% valid order reliability, analyzed ₹61.75M in revenue across 67K orders, and delivered actionable insights on payment methods, product performance, and geographic distribution.

Dataset Description:

The dataset consists of **100,000 e-commerce sales records**, including order, customer, product, seller, and geographic attributes. The data was cleaned, validated, and standardized using MySQL before being imported into Power BI for analysis and visualization.

Key Attributes in the Dataset Include:

- Order Details – OrderID, OrderDate, OrderStatus, Quantity, TotalAmount, ShippingCost
- Customer Info – CustomerID, CustomerName, City, State, Country
- Product Data – ProductID, ProductName, Category, Brand, UnitPrice
- Seller Metrics – SellerID, Brand, Country-level sales presence
- Data Quality Flags – is_valid_location, category_reliability

Data Preprocessing & Transformation:

- Performed schema validation, data type standardization, and data quality checks in MySQL
- Resolved city–country and product–category inconsistencies using mapping and lookup logic
- Aggregated revenue and order volume across year, quarter, month, and daily levels
- Loaded the transformed dataset into Power BI and developed DAX measures for KPI tracking and trend analysis
- Implemented interactive slicers for Country, Category, and Payment Method to enable dynamic filtering

Tools Used and Their Role in the Project:

- **MySQL** – Data cleaning, validation, transformation, KPI computation, and aggregation
- **Power BI** – Dashboard development, DAX-based analysis, and data visualization
- **Power BI Service** – Dashboard publishing, sharing, and stakeholder access

Power BI Dashboard Overview:

A single-page, KPI-driven analytics dashboard built on MySQL-processed e-commerce sales data, designed to deliver insights on trends, segmentation, operational performance, and geographic distribution.

Power BI Dashboard Includes:

- **KPI Cards** – Revenue, Orders, Average Order Value (AOV), Customers, Valid Order Percentage
- **Trend Analysis** – Monthly revenue trend using a continuous time-series line chart
- **Segmentation Analysis** – Revenue breakdown by Category, Payment Method, Country, and Order Status
- **Interactive Controls** – Slicers for Category, Country, and Payment Method for dynamic filtering
- **Operational Analysis** – Visualization of revenue impact across different order lifecycle stages

Key Insights & Findings:

- Revenue remains consistent across years with identifiable seasonal peaks
- Digital payment methods contribute the majority of revenue, indicating a shift from cash-based transactions
- Electronics and Home categories generate the highest revenue contribution
- Sales are distributed across five countries, highlighting geographic market presence
- Approximately 30% of records contained data inconsistencies, which were resolved in SQL, resulting in 70.1% valid order reliability

Business Recommendations:

- Implement automated data validation checks for city–country and product–category mappings to prevent inconsistencies at the source
- Focus on high-revenue categories (Electronics and Home) through targeted campaigns and bundle strategies to maximize profitability
- Promote digital payment methods through incentives such as cashback and seamless checkout, as they contribute the majority of revenue

- Monitor seller contribution to avoid over-dependence on a limited set of sellers and improve marketplace balance

Limitations of the Project:

- Dataset contained synthetic inconsistencies, requiring assumption-based corrections rather than real-world validation
- Analysis is based on historical data and does not support real-time insights
- Single-page dashboard limits detailed drill-down and hierarchical analysis
- Geographic insights may vary slightly due to data normalization using mapping logic

Future Scope:

- Integrate with a live MySQL database and enable scheduled refresh in Power BI Service for real-time analytics
- Incorporate forecasting models and anomaly detection for advanced decision support
- Implement drill-through, trend decomposition, and cohort analysis for deeper customer insights
- Expand into a multi-page BI report with operational monitoring and AI-driven recommendations

Conclusion:

This project demonstrates an end-to-end business intelligence workflow, transforming an inconsistent e-commerce dataset into reliable analytical insights through SQL-based data cleaning and KPI development. The dashboard provides a clear view of sales performance, customer behavior, category contribution, seller distribution, and geographic revenue trends, making it a strong portfolio project for data analyst roles.